

EX-12 Write the Python program for Tic Tac Toe Game

AIM

To develop a Python program to implement the **Tic Tac Toe Game** for two players, allowing them to play alternately until one player wins or the game ends in a draw.

ALGORITHM

Step 1: Start the program.

Step 2: Create a 3×3 game board initialized with empty spaces.

Step 3: Set Player 1 as **X** and Player 2 as **O**.

Step 4: Display the game board.

Step 5: Ask the current player to enter the row and column position.

Step 6: Check whether the selected position is empty.

Step 7: If the position is valid, place the player's symbol on the board.

Step 8: Check whether the current player has won by completing any row, column, or diagonal.

Step 9: If a player wins, display the winner and stop the game.

Step 10: If all positions are filled and no player wins, declare the game as a draw.

Step 11: Otherwise, switch to the next player and repeat from Step 4.

Step 12: Stop the program.

PROGRAM CODE

```
board = [' ' for _ in range(9)]
```

```
def display():
```

```
    print()
```

```
    print(board[0], "|", board[1], "|", board[2])
```

```
    print("---+---+---")
```

```
print(board[3], "|", board[4], "|", board[5])
print("---+---+---")
print(board[6], "|", board[7], "|", board[8])
print()
```

```
def winner(player):
    win = [
        [0,1,2],[3,4,5],[6,7,8],
        [0,3,6],[1,4,7],[2,5,8],
        [0,4,8],[2,4,6]
    ]
    for x in win:
        if board[x[0]] == board[x[1]] == board[x[2]] == player:
            return True
    return False
```

```
player = 'X'
```

```
for turn in range(9):
    display()
```

```
pos = int(input(f"Player {player}, Enter position (1-9): ")) - 1
```

```
if board[pos] == '':
```

```
board[pos] = player
```

```
if winner(player):
```

```
    display()
```

```
    print("Player", player, "Wins!")
```

```
    break
```

```
player = 'O' if player == 'X' else 'X'
```

```
else:
```

```
    print("Position already occupied!")
```

```
else:
```

```
    display()
```

```
    print("Match Draw!")
```

OUTPUT

```
  | X | X
--+---+--
O | O |
--+---+--
  |   |
```

```
Player X, Enter position (1-9): 2
Position already occupied!
```

```
  | X | X
--+---+--
O | O |
--+---+--
  |   |
```

```
Player X, Enter position (1-9): 6
```

```
  | X | X
--+---+--
O | O | X
--+---+--
  |   |
```

```
Player O, Enter position (1-9): 7
```

```
  | X | X
--+---+--
O | O | X
--+---+--
O |   |
```

```
Player X, Enter position (1-9): 8
```

```
  | X | X
--+---+--
O | O | X
--+---+--
O | X |
```

```
Match Draw!
```

RESULT

The Python program successfully implements the **Tic Tac Toe Game**. Two players can play alternately, and the program correctly identifies the winner or declares the match as a draw.